

CBSE Class 12 Pe — Half-Yearly Predicted Paper (Set 3 of 10)

SECTION A: Objective Type Questions (18 Marks: 18 x 1)

All questions are compulsory. Consists of MCQs and Assertion-Reason questions.

Q1. In a knock-out tournament, if the total number of byes (N_b) is an odd number, what is the formula to calculate the number of byes in the lower half of the fixture? (A) $\frac{N_b-1}{2}$ (B) $\frac{N_b+1}{2}$ (C) $\frac{N+1}{2}$ (D) $\frac{N-1}{2}$ Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High**

Q2. Which of the following conditions is an eating disorder characterized by a cycle of binge eating followed by extreme purging (e.g., self-induced vomiting)? (A) Amenorrhea (B) Anorexia Nervosa (C) Bulimia Nervosa (D) Osteoporosis Unit 2: Children & Women in Sports | [PYQ] | Tag: Very High**

Q3. Which of the following yoga asanas is famously known as the "Cow Face Pose"? (A) Gomukhasana (B) Bhujangasana (C) Halasana (D) Vajrasana Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: High**

Q4. The first Special Olympics International Summer Games were held in the year 1968 in which city? (A) London, UK (B) Athens, Greece (C) Chicago, USA (D) Rome, Italy Unit 4: Physical Education & Sports for CWSN | [Practice] | Tag: Medium**

Q5. Goiter is a condition characterized by the swelling of the neck resulting from the enlargement of the thyroid gland. It is primarily caused by the deficiency of which mineral? (A) Iron (B) Calcium (C) Iodine (D) Sodium Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High**

Q6. In the Rikli & Jones Senior Citizen Fitness Test, the 6-Minute Walk Test is administered specifically to measure: (A) Lower body flexibility (B) Upper body strength (C) Cardiovascular/Aerobic endurance (D) Agility and dynamic balance Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High**

Q7. What is the normal resting heart rate of a healthy adult human? (A) 60 beats/min (B) 72 beats/min (C) 90 beats/min (D) 100 beats/min Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: High**

Q8. The anatomical movement that involves moving a body part toward the imaginary midline of the body is known as: (A) Abduction (B) Adduction (C) Flexion (D) Extension Unit 8: Biomechanics and Sports | [PYQ 2022] | Tag: Very High**

Q9. Which committee is responsible for formally welcoming the chief guests, spectators, and participating teams during a sporting event? (A) Technical Committee (B) Reception Committee (C) Marketing Committee (D) Transport Committee Unit 1: Management of Sporting Events | [PYQ] | Tag: High**

Q10. Medically, the postural deformity known as 'Genu Varum' refers to: (A) Knock Knees (B) Flat Foot (C) Bow Legs (D) Round Shoulders Unit 2: Children & Women in Sports | [Practice] |

Tag: High**

Q11. *Nadi Shodhana* and *Suryabhedan* are specific types of: (A) Asanas (B) Pranayamas (C) Mudras (D) Kriyas *Unit 3: Yoga as Preventive Measure for Lifestyle Disease* | [PYQ] | Tag: Very High**

Q12. What does the acronym ODD stand for in the context of behavioral disorders in children? (A) Obsessive Defect Disability (B) Oppositional Defiant Disorder (C) Opposite Different Disorder (D) Occasional Defiant Disability *Unit 4: Physical Education & Sports for CWSN* | [PYQ] | Tag: Very High**

Q13. Water is an essential non-nutritive component of the human diet. It makes up approximately what percentage of the total human body weight? (A) 30% - 40% (B) 45% - 55% (C) 60% - 70% (D) 80% - 90% *Unit 5: Sports & Nutrition* | [PYQ] | Tag: High**

Q14. In the SAI Khelo India Fitness Test battery for the 9-18 years age group, which test is used to measure a student's lower back and hamstring flexibility? (A) Sit and Reach Test (B) Partial Curl-up (C) 50m Dash (D) Flamingo Balance Test *Unit 6: Test and Measurement in Sports* | [PYQ] | Tag: Very High**

Q15. An irregular, jagged tear of the skin caused by a blunt force or rough impact (like falling on an uneven field) is classified as a: (A) Laceration (B) Incision (C) Contusion (D) Strain *Unit 7: Physiology & Injuries in Sport* | [PYQ] | Tag: Very High**

Q16. Assertion (A): A lighter tennis ball accelerates much faster than a heavy shot put when thrown with the exact same amount of force. **Reason (R):** Newton's Second Law of Motion states that acceleration is directly proportional to force and inversely proportional to the mass of the object. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 8: Biomechanics and Sports* | [PYQ] | Tag: Very High**

Q17. Assertion (A): Extramural tournaments help in finding and exposing hidden athletic talents by providing a broader competitive platform. **Reason (R):** Extramural competitions are those played strictly outside the walls of an institution involving teams from various different schools. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 1: Management of Sporting Events* | [Practice] | Tag: High**

Q18. Assertion (A): The Back Scratch Test is used in senior citizens to measure their lower body flexibility. **Reason (R):** Flexibility tends to naturally decrease as a physiological change due to aging. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 6: Test and Measurement in Sports* | [PYQ] | Tag: Very High**

SECTION B: Very Short Answer Type (10 Marks: Attempt any 5 out of 6 x 2)

Word limit: 60-90 words.

- Q19.** State any two main objectives of organizing community sports programs like 'Run for Fun'. *Unit 1: Management of Sporting Events | [PYQ] | Tag: High***
- Q20.** Enlist any two common causes that lead to the spinal deformity known as 'Scoliosis'. *Unit 2: Children & Women in Sports | [PYQ] | Tag: Very High***
- Q21.** Mention any two contraindications (who should avoid it) for performing *Halasana* (Plough Pose). *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: High***
- Q22.** What do you understand by the term 'Disability Etiquette'? Give one appropriate example. *Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: High***
- Q23.** Define 'Food Intolerance'. State any one common physical symptom associated with it. *Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High***
- Q24.** Explain the term 'Greenstick Fracture' in the context of sports injuries. *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High***
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SECTION C: Short Answer Type (15 Marks: Attempt any 5 out of 6 x 3)

Word limit: 100-150 words.

- Q25.** Outline the pre-tournament responsibilities of the Technical Committee in organizing a sports event. *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***
- Q26.** Differentiate clearly between 'Menarche' and 'Menstrual Dysfunction' in female athletes. *Unit 2: Children & Women in Sports | [PYQ] | Tag: High***
- Q27.** Discuss the physiological roles of Roughage (fiber) and Water as critical non-nutritive components of a balanced diet. *Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High***
- Q28.** Describe the purpose, equipment required, and the administration procedure for the Chair Stand Test for senior citizens. *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***
- Q29.** What is a 'Joint Dislocation'? How should it be managed immediately on the sports field? *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: High***
- Q30.** Differentiate between the anatomical movements of 'Flexion' and 'Extension' by providing suitable examples from sports. *Unit 8: Biomechanics and Sports | [PYQ 2022] | Tag: Very High***
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SECTION D: Case-Study Based Questions (12 Marks: 3 x 4)

All 3 questions are compulsory. Internal choice is provided in one sub-part of each.

- Q31.** The sports captain of a school is learning how to draw fixtures using the League format for an upcoming inter-house volleyball tournament. The Physical Education teacher asks him to use the 'Staircase Method' to draw a fixture for 7 teams, as it is considered one of the easiest methods where no byes are required. **(i)** What will be the total number of matches

played in this tournament? **(ii)** Is there any 'Bye' given in the staircase method of a league tournament? (Yes/No) **(iii)** How many total rounds will be there in this fixture? **(iv)** Which type of tournament format is considered the best to judge the true winner? **OR** Write the mathematical formula used to calculate the number of matches in a Single League tournament. *Unit 1: Management of Sporting Events | [Practice] | Tag: Very High***

Q32. A health check-up camp was organized in a school for class 12 students. The doctor recorded their height and weight to calculate their Body Mass Index (BMI). The final report showed: 15% of the students had a BMI < 18.5; 50% were between 18.5 — 24.9; 25% were between 25 — 29.9; and 10% had a BMI > 30. **(i)** Which WHO weight category do the 15% of students with a BMI < 18.5 fall into? **(ii)** State the standard mathematical formula used to calculate BMI. **(iii)** The 10% of students with a BMI > 30 are suffering from which specific lifestyle disease? **(iv)** Mention one major health risk associated with a BMI > 30. **OR** Name one macro-nutrient that the obese students should reduce in their diet to help control their weight. *Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High***

Q33. During a swimming competition, an athlete dives off the starting block, gliding through the air, and then swims using the freestyle stroke. The coach later explains the biomechanics behind the swim: "The water pushes you forward only because you are pulling and pushing the water backward with your hands and feet." **(i)** The statement made by the coach is a practical application of which of Newton's Laws of Motion? **(ii)** When the swimmer is actively swimming across the pool maintaining a steady pace, their body is in which state of equilibrium? (Static or Dynamic) **(iii)** The resistance offered by the water that opposes the swimmer's forward motion is known as _ friction. **(iv)** Briefly define the term 'Centre of Gravity'. **OR** Newton's First Law of Motion is also universally known as the Law of _ . *Unit 8: Biomechanics and Sports | [PYQ] | Tag: High***

SECTION E: Long Answer Type (15 Marks: Attempt any 3 out of 4 x 5)

Word limit: 200-300 words.

Q34. What is Asthma? Enlist any four Yoga Asanas recommended for its prevention. Thoroughly describe the step-by-step procedure and two benefits of Gomukhasana (Cow Face Pose). *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: Very High***

Q35. Clearly define the terms 'Disability' and 'Disorder'. Discuss in detail any four effective strategies that can be implemented by physical education teachers to make physical activities and sports accessible for Children With Special Needs (CWSN). *Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: Very High***

Q36. The Sports Authority of India (SAI) Khelo India Fitness Test battery tracks the physical fitness of school children. Elaborate in detail on the three specific test items prescribed for the age group of 5-8 years (Classes 1 to 3), including their purpose and administration procedure. *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q37. Define 'Bone Fractures' in sports. Classify and thoroughly explain any four types of bone fractures (e.g., Simple, Compound, Comminuted, Greenstick, Transverse) outlining how they occur. *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: High***

Answer Key

Q1. (B) $\frac{N_b+1}{2}$ **Q2.** (C) Bulimia Nervosa **Q3.** (A) Gomukhasana **Q4.** (C) Chicago, USA **Q5.** (C) Iodine **Q6.** (C) Cardiovascular/Aerobic endurance **Q7.** (B) 72 beats/min **Q8.** (B) Adduction **Q9.** (B) Reception Committee **Q10.** (C) Bow Legs **Q11.** (B) Pranayamas **Q12.** (B) Oppositional Defiant Disorder **Q13.** (C) 60% - 70% **Q14.** (A) Sit and Reach Test **Q15.** (A) Laceration **Q16.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q17.** (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (Reason does not directly explain *why* it helps find hidden talent, though both statements are factually correct). **Q18.** (D) (A) is false, but (R) is true. (Back scratch test measures *upper body* flexibility, not lower body).

Q19. 1. To promote health, fitness, and an active lifestyle among the masses. 2. To provide recreation and relieve stress without the pressure of intense competition. **Q20.** 1. Congenital defects or uneven leg lengths. 2. Carrying heavy backpacks on one shoulder consistently. **Q21.** 1. Individuals suffering from severe neck or spine injuries (cervical spondylosis). 2. Individuals with high blood pressure or pregnant women should avoid it. **Q22.** Disability Etiquette refers to respectful behavioral guidelines and manners when interacting with people with disabilities. **Example:** Always speaking directly to the person with a disability rather than addressing their interpreter or companion. **Q23. Food Intolerance** is the inability of the digestive system to properly break down and digest certain foods (e.g., lactose), causing discomfort. **Symptom:** Nausea, bloating, or stomach ache. **Q24. A Greenstick Fracture** is an incomplete fracture where the soft bone bends and cracks but does not break completely into separate pieces. It is most commonly seen in young children because their bones are softer and more flexible.

Q25. The Technical Committee handles the sporting rules and field setup. **Pre-tournament responsibilities:** 1. Purchasing or arranging all necessary sports equipment. 2. Preparing and marking the playfields/courts according to standard measurements. 3. Drafting the rules, scheduling matches, and preparing the tournament fixtures. 4. Arranging qualified officials (referees, umpires). **Q26. Menarche** is a natural biological milestone; it is the very first menstrual cycle or onset of menstruation in a young female. **Menstrual Dysfunction**, however, is an abnormal condition where the menstrual cycle becomes irregular, extremely painful (dysmenorrhea), or stops entirely (amenorrhea) often due to overtraining or poor diet. **Q27. Roughage (Fiber):** It adds bulk to the diet, making the stomach feel full without adding calories. It prevents constipation and ensures the smooth functioning of the digestive tract. **Water:** It makes up about 60% of the body. It regulates body temperature (thermoregulation via sweating), transports nutrients to cells, and flushes out toxins, though it provides zero calories. **Q28. Purpose:** To assess lower body strength in senior citizens, essential for tasks like climbing stairs. **Equipment:** A straight-back chair (approx. 17 inches

high) and a stopwatch. **Procedure:** The participant sits in the middle of the chair, feet flat on the floor, arms crossed over the chest. On the "go" command, they rise to a full stand and sit back down. The score is the maximum number of complete stands executed in 30 seconds.

Q29. A Joint Dislocation is a severe bone injury where the adjoining bones are displaced from their normal anatomical position in the joint (e.g., shoulder dislocation). **Management:** Do not attempt to force the bone back into place. Immobilize the joint immediately using a splint or sling, apply ice to reduce swelling, and seek urgent professional medical help. **Q30. Flexion** is bending a joint so that the angle between the bones decreases (e.g., bending the elbow during the upward phase of a bicep curl, or bending the knee to prepare for a jump). **Extension** is straightening a joint so the angle increases (e.g., straightening the arm fully while releasing a javelin or executing a shot put throw).

Q31. (i) Matches = $\frac{N(N-1)}{2} = \frac{7(6)}{2} = 21$ matches. (ii) No. (iii) Rounds = $N - 1 = 7 - 1 = 6$ rounds. (iv) League Tournament (Round Robin). **OR** $\frac{N(N-1)}{2}$. **Q32.** (i) Underweight. (ii) $\frac{\text{Weight (in kg)}}{\text{Height (in meters)}^2}$. (iii) Obesity. (iv) High risk of heart diseases / Type 2 Diabetes / Hypertension. **OR** Carbohydrates or Fats. **Q33.** (i) Newton's Third Law of Motion (Law of Action and Reaction). (ii) Dynamic Equilibrium. (iii) Fluid friction. (iv) The Centre of Gravity is the imaginary point around which the entire weight of the body is evenly balanced. **OR** Law of Inertia.

Q34. Asthma is a chronic respiratory disease where the airways become inflamed, narrow, and produce extra mucus, making breathing difficult. **Asanas for Prevention:** Gomukhasana, Chakrasana, Matsyasana, Bhujangasana. **Gomukhasana Procedure:** Sit with legs extended. Bend the left leg and place the left heel under the right hip. Bend the right leg over the left knee, placing the right heel near the left hip. Raise the right arm upwards, bend it at the elbow, and take it behind the back. Take the left arm behind the back from the bottom and interlock the fingers of both hands. Keep the spine and head straight. **Benefits:** 1. Expands the chest, improving lung capacity (relieves asthma). 2. Cures postural deformities like round shoulders and stiff back. **Q35. Disability:** A physical or mental restriction that limits a person's ability to perform standard daily activities (e.g., blindness, amputation). **Disorder:** A disruption of normal physiological or psychological functions (e.g., ADHD, OCD). **Strategies for Inclusion:** 1. **Adapted Equipment:** Using lighter bats, balls with internal bells (for the visually impaired), or lower basketball hoops. 2. **Rule Modification:** Giving extra time, reducing court size, or allowing extra attempts. 3. **Accessible Environment:** Ensuring ramps, wheelchair-friendly courts, and smooth surfaces to prevent falls. 4. **Peer Buddy System:** Pairing a mainstream student with a CWSN student to assist and encourage them during play. **Q36.** The SAI Khelo India battery for 5-8 years tracks basic motor skills and body composition. **1. BMI (Body Mass Index): Purpose:** To measure body composition and categorize weight. **Administration:** Height is measured in meters and weight in kg. BMI is calculated using the formula Weight/Height^2 . **2. Flamingo Balance Test: Purpose:** To measure static balance. **Administration:** The child stands barefoot on a wooden beam on one leg, bending the free leg and holding it close to the buttocks. Time starts when the instructor lets go. Falls in 60 seconds are counted. **3. Plate Tapping Test: Purpose:** To measure

coordination and speed of upper limbs. *Administration:* Two yellow discs and a center rectangle are placed on a table. The child taps the two discs alternately over the rectangle. The time taken to complete 25 cycles (50 taps) is recorded. **Q37. A Bone Fracture** is a partial or complete break in the continuity of a bone caused by high impact or stress. **1. Simple Fracture:** The bone breaks completely, but the broken ends do not pierce the skin. **2. Compound (Open) Fracture:** A severe fracture where the broken bone violently pierces through the skin, exposing the bone and increasing infection risk. **3. Comminuted Fracture:** The bone is crushed or shattered into three or more pieces, usually due to high-impact trauma like a heavy tackle. **4. Greenstick Fracture:** An incomplete break where the bone bends and cracks, commonly seen in young children due to softer bones. **5. Transverse Fracture:** A straight break directly across the horizontal axis of the bone shaft.

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.